

BAKEFLOW

Day 1 — Product Requirements Document

Bakery Commerce + POS + Inventory Management Platform

Working idea: BakeFlow is one platform for a bakery's online customers and daily operations. Customers can order online, while bakery staff can handle walk-in/POS orders, payments, billing, inventory, kitchen work, fulfilment tracking and support.

This PRD is written in simple language and will act as our product source of truth. Database schemas, APIs and code will be designed after the requirements and workflows are finalized.

Product	BakeFlow
Core idea	Bakery Commerce + POS + Inventory
Main users	Customer, Staff, Kitchen Staff, Support, Admin
Goal	A realistic, deployable, hiring-ready full-stack business application

1. Product Overview

- BakeFlow is a digital system for a bakery that brings online ordering and internal bakery operations together.
- The goal is not to build only a pretty bakery website. The system should help the business receive orders, take payments, prepare orders, manage stock, generate bills, track fulfilment, handle refunds and understand sales.
- The same order engine should support both online customer orders and offline/walk-in orders created through the bakery POS.

2. Problem We Are Solving

- Online orders, walk-in orders, billing and inventory can become separate processes.
- Staff may have to manually check stock before accepting an order.
- Stock can become inaccurate when sales, cancellations, waste and manual adjustments are not recorded properly.
- Customers may not know what is happening with an order after payment.
- Refund and complaint handling can be difficult when order and payment information are disconnected.
- The owner may not have one place to see products, orders, revenue, customers, staff and inventory.
- Adding a new dessert or changing a product should not require a developer to change code.

3. Product Goals

- Make online ordering simple for customers.
- Give staff a fast POS flow for walk-in customers.
- Allow authorized admin users to add, edit, activate/deactivate and manage products directly from the admin panel.
- Connect online and POS orders to the same core order and inventory logic.
- Maintain reliable inventory and prevent invalid stock deductions.
- Handle payments, invoices, cancellations and refunds in a traceable way.
- Give kitchen staff a simple preparation queue and fulfilment workflow.
- Let customers see meaningful, real-time order progress.
- Provide structured customer support for refund, payment and order issues.
- Give admins useful sales and operational analytics.
- Use professional engineering practices so the project can be discussed confidently in interviews.

4. Users & Roles

Customer

- Create account and manage profile.
- Browse/search products and add them to cart.
- Checkout, choose delivery/pickup and pay.
- Track order progress through each fulfilment stage.
- View previous orders and invoices.
- Cancel eligible orders and raise refund/support requests.

Staff / Cashier

- Create walk-in orders using POS.
- Search products and add quantities quickly.
- Record cash/UPI payments.

- Generate and print bills.
- View and process relevant orders.
- Update permitted fulfilment statuses.

Kitchen Staff

- See newly confirmed orders.
- Start cooking/preparation.
- Mark orders as packed when ready.
- Update only the operational statuses allowed for the kitchen.

Support Agent

- View support tickets.
- See related order and payment information.
- Handle payment, cancellation, delivery, wrong-item, invoice and refund queries.
- Update ticket status and resolution.

Admin / Bakery Manager

- Add new products/desserts without code changes.
- Edit product name, description, price, image, category, availability and other configured details.
- Activate, deactivate, mark out of stock, or discontinue products.
- Manage categories and product catalogue.
- Manage inventory, orders, customers, staff, coupons and support.
- Review payments/refunds and sales analytics.
- Control access by role.

5. Main Modules

- Authentication and user profiles
- Dynamic product catalogue and category management
- Cart and checkout
- Order management and fulfilment tracking
- Bakery POS for walk-in orders
- Payment processing
- Billing and invoice generation
- Inventory and inventory transactions
- Kitchen order management
- Customer support and refund handling
- Admin dashboard and analytics
- Notifications

6. Product Management — Admin

- Admin can create a new bakery product/dessert from the admin panel instead of hardcoding products in the frontend.
- Product form can include name, description, category, price, images, availability, stock-related information, preparation time, ingredients/options and tags where applicable.
- Admin can edit an existing product.
- Admin can activate/deactivate a product or mark it out of stock.
- Only active/available products should appear for customers.
- When an admin saves a new active product, it should automatically become available on the customer-facing product catalogue through the product API.

- Product images should be stored using an image-storage service rather than inside the database.

Example flow

Admin → Products → Add New Product → Fill details → Upload image → Save → Backend validates → Database stores product → Product API returns it → Active product appears on customer product page.

7. Main User Flows

Customer online order

Browse products → Product details → Add to cart → Cart → Checkout → Address + delivery/pickup → Payment → Payment verification → Order confirmed → Cooking/Preparing → Packing → Out for Delivery or Ready for Pickup → Delivered/Picked Up → Invoice.

Staff / POS order

Staff opens POS → Selects products → Adds customer details if needed → Applies allowed discount → Records cash/UPI payment → Order created → Bill generated/printed → Order enters the same kitchen/fulfilment workflow.

Cancellation / refund

Customer requests cancellation/refund → System checks order/payment state and rules → Support/authorized staff reviews → Approved when eligible → Inventory restored where applicable → Refund status updated → Customer notified → Ticket resolved.

Inventory

Stock is added/adjusted → Online or POS sale reduces stock → Eligible cancellation/return can restore stock → Waste/manual adjustment is recorded → Low-stock condition can trigger an alert.

Customer support

Customer opens 24×7 support → AI assistant handles common questions/order information → Unresolved issue becomes a ticket → Support reviews order/payment → Issue resolved → Ticket closed.

8. Order Fulfilment & Status Tracking

- Every order has a controlled status that represents where it currently is in the bakery fulfilment process.
- Customer-facing status timeline should clearly show the progress of the order.
- Typical delivery flow: PLACED → CONFIRMED → PREPARING/COOKING → PACKED → OUT_FOR_DELIVERY → DELIVERED.
- For pickup orders: PLACED → CONFIRMED → PREPARING/COOKING → PACKED/READY_FOR_PICKUP → PICKED_UP.
- Alternative states such as CANCELLED, PAYMENT_FAILED, REFUND_PENDING and REFUNDED are handled separately from the normal fulfilment path.
- Each status change should be recorded in order-status history with time and relevant actor where appropriate.
- Backend rules must control which status transitions are valid; staff should not be able to arbitrarily move an order from DELIVERED back to PREPARING.
- Customer notifications can be triggered when important status changes occur.

Example customer timeline

✓ Order Confirmed → ✓ Cooking/Preparing → ✓ Packing → ● Out for Delivery → ○ Delivered

Example internal workflow

Customer places order → Payment verified → Staff confirms → Kitchen prepares → Kitchen packs → Staff/Delivery updates out-for-delivery → Delivery completed.

9. V1 Scope — Must Have

- Customer registration/login and profile
- Dynamic product catalogue and admin product creation/editing
- Categories, search and product details
- Cart and checkout
- Delivery/pickup selection
- Razorpay test-mode payment
- Order creation, history, fulfilment tracking and eligible cancellation
- POS for walk-in orders
- Cash/UPI payment recording and invoice printing
- Product and inventory management
- Inventory transaction history and low-stock handling
- Kitchen order queue and status updates
- Customer support tickets and refund/payment queries
- Admin management and basic sales analytics
- Role-based access control
- Critical workflow testing and deployment

10. V2 / Nice to Have

- Richer AI support assistant
- Loyalty/reward points
- Subscription orders
- Demand forecasting
- Multi-branch bakery support
- WhatsApp/SMS notifications
- Advanced delivery routing

These are deliberately outside V1 so we finish a smaller number of complete workflows instead of creating an unfinished project with too many features.

11. Important Business Rules

- Backend is the final authority for price, discount, tax, stock and order totals.
- Customers can cancel only when the order is in an allowed state.
- Delivered orders cannot be cancelled through the normal customer flow.
- Inventory should not become negative because of simultaneous purchases.
- Online and POS orders should use the same core order rules wherever possible.
- Duplicate payment/webhook events must not create duplicate business actions.
- Only authorized roles can access admin, POS, kitchen, support and inventory operations.
- Refund handling must be connected to the original order and payment.
- Only valid order-status transitions are allowed.
- Every important status and inventory change should be traceable.
- Inactive or unavailable products must not be purchasable through the customer storefront.

12. Outside V1

- No multi-bakery marketplace.
- No separate mobile application.
- No full accounting/ERP system.
- No microservices just for complexity.
- No claim of actual human 24x7 support; the platform provides 24x7 self-service/AI access and ticket creation.
- No AI feature unless it solves a real support or operational problem.

13. Success Criteria

- A customer can complete an online order from product discovery to payment, fulfilment tracking and invoice.
- A staff member can create a walk-in order and print a bill through POS.
- Online and POS orders use the same core order logic and fulfilment statuses.
- An admin can add a new dessert/product and make it visible on the customer catalogue without changing code.
- Inventory changes correctly for sales, cancellations, waste and adjustments.
- Kitchen staff can see and update preparation/packing status.
- A customer can see meaningful order progress from confirmation through delivery/pickup.
- A customer can raise a support/refund query and see its status.
- Admins can manage core business data and see meaningful sales information.
- Critical APIs and business rules have automated tests.
- The application is deployed and documented with a working demo.

14. Hiring-Ready Engineering Direction

- Frontend: Next.js + TypeScript with a clean responsive interface.
- Backend: Node.js + Express + TypeScript with controllers, services, validation and middleware.
- Database: MongoDB + Mongoose with deliberate schema design and indexes.
- Authentication: secure authentication, HTTP-only cookies and bcrypt password hashing.
- Authorization: RBAC for Customer, Staff, Kitchen, Support and Admin.
- Payments: Razorpay test mode with server-side verification and webhooks.
- Testing: backend and frontend tests focused on critical business workflows.
- Documentation: ER diagram, architecture diagram, API documentation, README and demo video.

15. Final Product Definition

BakeFlow is not just a bakery website. It is a bakery commerce and operations platform where online customers and bakery staff use connected workflows for product management, ordering, payment, billing, inventory, kitchen preparation, fulfilment tracking and customer support. We will prioritize complete and reliable workflows over a large number of shallow features.